

Areas of regular n -gons in Euclidean geometry

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Abstract

This paper follows a very brief derivation of the formula for determining the area of any regular n -gon in Euclidean space. This derivation was entirely independent, conducted without referencing existing mathematical literature.

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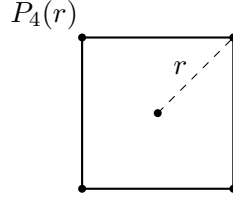
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1 Introduction

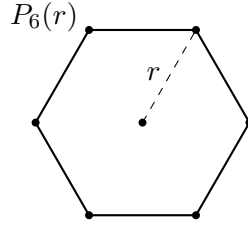
1.1 Regular n -gon objects

Throughout this paper, we will let $P_n(r) \subseteq \mathbb{R}^2 : n \in [3, \infty) \cap \mathbb{Z}, r \in [0, \infty)$, which will be centered at the origin, represent a regular n -gon (a regular polygon with n sides and n vertices) in Euclidean space (hence $n \geq 3$) with circumradius r .

For example, $P_4(r)$ represents the following regular polygon:



Likewise, $P_6(r)$ represents the following regular polygon:

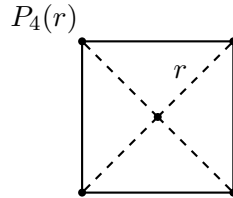


1.2 The A function

Let the set, $\mathcal{P} = \{P_n(r) : n \in [3, \infty) \cap \mathbb{Z}\}$ represent the set of all regular n -gons, where $r \in [0, \infty)$ is a variable parameter. We can now define the function $A : \mathcal{P} \rightarrow [0, \infty)$, which maps each polygon in $\mathcal{P}$ to its area. Thus, for a polygon $P_n(r) \in \mathcal{P}$, $A(P_n(r))$ denotes its area, with the dependence on n and r arising through the definition of $P_n(r)$.

1.3 Splitting n -gons into n congruent triangles

It is well known that a regular n -gon can be divided into n congruent triangles, each with two sides equal to the circumradius r and the remaining side equal to the side length of the n -gon. For example, $P_4(r)$ can be divided into 4 congruent triangles:



Hereafter, I will use this result without further justification.

2 Derivation

2.1 Deriving A

Consider the following equation:

$$z^n = 1 : z \in \mathbb{C}$$

$$\therefore z^n = \cos(2k\pi) + i \sin(2k\pi) : k \in \mathbb{Z}$$

Using De Moivre's theorem let's us determine the following:

$$z_k = \cos\left(\frac{2k\pi}{n}\right) + i \sin\left(\frac{2k\pi}{n}\right)$$

Then under the mapping

$$\phi : \mathbb{C} \rightarrow \mathbb{R}^2, \phi(a + bi) = (a, b),$$

we obtain

$$\phi(z_k) = \left(\cos\left(\frac{2k\pi}{n}\right), \sin\left(\frac{2k\pi}{n}\right) \right).$$

z_k are roots of unity, so that each $\phi(z_k) \in \mathbb{R}^2$ describes a vertex of a regular n -gon, with a radius of 1 (as the modulus of z_k is 1) centered at the origin, $P_n(1)$. The vertices of $P_n(r)$ can therefore be described by $\phi(rz_k)$, since multiplying z_k by r preserves its argument while scaling its modulus from 1 to r . We can immediately write that

$$\phi(rz_k) = \left(r \cos\left(\frac{2k\pi}{n}\right), r \sin\left(\frac{2k\pi}{n}\right) \right).$$

so that each $\phi(rz_k)$ is a vertex of $P_n(r)$. Each vertex, $\phi(rz_k)$, is at an anticlockwise angle of $\frac{2k\pi}{n}$ from the positive x -axis. Thus, $\phi(rz_{k+1})$, will be the subsequent anticlockwise vertex of the regular n -gon, $P_n(r)$, at an angle $\frac{2(k+1)\pi}{n}$ from the positive x -axis.

The apothem, a , is the perpendicular distance from the centre of a regular polygon to any of its sides. In a regular polygon, this perpendicular intersects each side at its midpoint. The midpoint of any side is the midpoint of the two vertices at either end. It follows that this perpendicular will intersect a side of $P_n(r)$ at the midpoint of the two adjacent vertices $\phi(rz_k)$ and $\phi(rz_{k+1})$. If M represents the midpoint of these two vertices, then, $a = |\vec{OM}|$. Using our expression for $\phi(rz_k)$, we can easily determine M :

$$M \left(r \left(\frac{\cos\left(\frac{2k\pi}{n}\right) + \cos\left(\frac{2(k+1)\pi}{n}\right)}{2} \right), r \left(\frac{\sin\left(\frac{2k\pi}{n}\right) + \sin\left(\frac{2(k+1)\pi}{n}\right)}{2} \right) \right)$$

Thus,

$$a = \sqrt{\left(r \left(\frac{\cos\left(\frac{2k\pi}{n}\right) + \cos\left(\frac{2(k+1)\pi}{n}\right)}{2} \right) \right)^2 + \left(r \left(\frac{\sin\left(\frac{2k\pi}{n}\right) + \sin\left(\frac{2(k+1)\pi}{n}\right)}{2} \right) \right)^2},$$

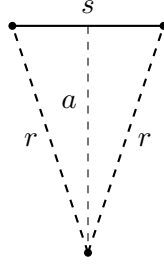
$$\therefore a = r \sqrt{\frac{2 + 2 \left(\cos\left(\frac{2k\pi}{n}\right) \cos\left(\frac{2(k+1)\pi}{n}\right) + \sin\left(\frac{2k\pi}{n}\right) \sin\left(\frac{2(k+1)\pi}{n}\right) \right)}{4}}.$$

Using the addition formula for cosine, we can determine that

$$a = r \sqrt{\frac{2 + 2 \cos \left(\frac{2k\pi}{n} - \frac{2(k+1)\pi}{n} \right)}{4}},$$

$$\therefore a = r \sqrt{\frac{2 - 2 \cos \left(\frac{2\pi}{n} \right)}{4}}.$$

When we split $P_n(r)$ into n congruent triangles, a will be the height of these triangles, as shown in the following diagram by the dashed line:



Where s represents the length of each side of $P_n(r)$. Thus,

$$\left(\frac{1}{2}s \right)^2 + a^2 = r^2,$$

$$\therefore s = 2\sqrt{r^2 - a^2}.$$

Using our expression for a , we obtain that

$$s = 2\sqrt{r^2 \left(1 - \frac{2 - 2 \cos \left(\frac{2\pi}{n} \right)}{4} \right)},$$

$$\therefore s = r \sqrt{2 + 2 \cos \left(\frac{2\pi}{n} \right)}.$$

Clearly, the area of one of these triangles can be given by

$$\frac{1}{2}sa = \frac{1}{2}r^2 \sqrt{2 + 2 \cos \left(\frac{2\pi}{n} \right)} \sqrt{\frac{2 - 2 \cos \left(\frac{2\pi}{n} \right)}{4}},$$

$$\therefore \frac{1}{2}sa = \frac{1}{2}r^2 \sqrt{1 - \cos^2 \left(\frac{2\pi}{n} \right)} = \frac{1}{2}r^2 \sin \left(\frac{2\pi}{n} \right).$$

If we have split $P_n(r)$ into n triangles with area $\frac{1}{2}sa$, then $P_n(r)$ will have an area of $\frac{n}{2}sa$ (the sum of the areas of these n triangles). Therefore, we can determine a final expression for our function, A :

$$A : \mathcal{P} \rightarrow [0, \infty), \quad A(P_n(r)) = \frac{n}{2}r^2 \sin \left(\frac{2\pi}{n} \right),$$

where,

- $\mathcal{P} = \{P_n(r) : n \in [3, \infty) \cap \mathbb{Z}\},$
- $r \in [0, \infty).$

This now allows us to determine the area of a regular n -gon with radius r .

2.2 s rather than r

If know the length of the sides, s , of $P_n(r)$, then we can determine the area of $P_n(r)$ using a slight variation of our function, A . We have that

$$A(P_n(r)) = \frac{n}{2} r^2 \sin\left(\frac{2\pi}{n}\right),$$

and

$$s = r \sqrt{2 + 2 \cos\left(\frac{2\pi}{n}\right)}.$$

Thus,

$$r^2 = \frac{s^2}{2 + 2 \cos\left(\frac{2\pi}{n}\right)}.$$

Therefore,

$$A(P_n(r)) = \frac{ns^2 \sin\left(\frac{2\pi}{n}\right)}{4 + 4 \cos\left(\frac{2\pi}{n}\right)},$$

where,

- $s \in [0, \infty)$.

This allows us to determine the area of a regular n -gon, when we only have the side length, s , without having to determine radius, r , beforehand.

2.3 Deriving the equation for the area of a circle from A

For $P_n(r)$, as $n \rightarrow \infty$, the vertices of $P_n(r)$ become arbitrarily dense on the circle of radius, r , centered at the origin. Consequently, $P_n(r)$ converges to this circle in the limit. Therefore,

$$\lim_{n \rightarrow \infty} A(P_n(r))$$

should converge to the equation for the area of a circle with radius r . We can observe that

$$\lim_{n \rightarrow \infty} A(P_n(r)) = \lim_{n \rightarrow \infty} \left(\frac{n}{2} r^2 \sin\left(\frac{2\pi}{n}\right) \right) = r^2 \lim_{n \rightarrow \infty} \left(\frac{n}{2} \sin\left(\frac{2\pi}{n}\right) \right).$$

Using the Taylor expansion of $\sin(x)$, centered at $x = 0$, lets us determine that

$$\lim_{n \rightarrow \infty} \left(\frac{n}{2} \sin\left(\frac{2\pi}{n}\right) \right) = \lim_{n \rightarrow \infty} \left(\frac{n}{2} \left(\frac{2\pi}{n} - \frac{4\pi^3}{3n^3} + \frac{4\pi^5}{15n^5} - \dots \right) \right) = \lim_{n \rightarrow \infty} \left(\pi - \frac{2\pi^3}{3n^2} + \frac{2\pi^5}{15n^4} - \dots \right) = \pi.$$

Therefore,

$$\lim_{n \rightarrow \infty} \left(\frac{n}{2} \sin\left(\frac{2\pi}{n}\right) \right) = \pi.$$

So,

$$\lim_{n \rightarrow \infty} A(P_n(r)) = r^2 \lim_{n \rightarrow \infty} \left(\frac{n}{2} \sin\left(\frac{2\pi}{n}\right) \right) = \pi r^2,$$

giving us the equation for the area of a circle, as expected.